Md Arif Hossain

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Profile Summary

I completed my B.Sc. in Mathematics from Shahjalal University of Science and Technology and currently work as a Business Intelligence Analyst in the manufacturing industry. My background in mathematics, computational research, Operations Research, and industrial analytics has shaped my interest in mathematical optimisation and data-driven decision-making. Through research in computational mathematics, independent work on facility-location and routing optimisation, and professional experience with manufacturing capacity, resource allocation, and operational decision support, I have become particularly interested in developing mathematical and computational methods for large-scale manufacturing, supply-chain, and operational systems.

Research Interests

Operations Research; Mathematical Optimisation; Mixed-Integer Programming; Large-Scale Systems Optimisation; Supply Chain Optimisation; Manufacturing Systems; Facility Location and Routing; Data-Driven Decision-Making; Computational Mathematics.

Education

B.Sc. in Mathematics 2019–2024
Shahjalal University of Science and Technology (SUST) *Sylhet, Bangladesh*
CGPA: 3.28/4.00

Research Experience

Research Student September 2023–May 2024
SUST Research Center *Sylhet, Bangladesh*
Implementation of Matrix Methods for Combinatorial Aspects of Posets and Lattices (Project No. PS/2023/1/19)
Project Investigator: **Dr. Mohammad Salah Uddin**, Associate Professor, Department of Mathematics, SUST

- Investigated matrix-based representations of partially ordered sets and lattices to analyze combinatorial structures.
- Developed and implemented computational methods in MATLAB for matrix representations and combinatorial analysis.
- Studied mathematical and computational properties of lattice-theoretic systems through matrix-based methods.

Undergraduate Thesis

On the Determinant, Inverse, and Eigenvalues of Infinite Matrices
Numerical and Computational Mathematics

- Studied determinants, inverses, eigenvalues, matrix logarithms, convergence, and series-expansion methods for infinite matrices.
- Developed computational approaches for numerical approximation of matrix-related quantities.

Selected Projects in Operations Research and Decision Analytics

Geographic Facility Location and Metric Routing Optimisation in Poland
Python | Gurobi | PuLP | SciPy/HiGHS | NetworkX

- Formulated a minimum set-covering facility-location model to identify store locations satisfying a 50-km geographic-coverage requirement across populated locations in Poland.
- Developed exact metric Traveling Salesman Problem formulations using degree and subtour-elimination constraints and compared explicit exponential constraints with iterative constraint-generation approaches.
- Implemented Christofides' approximation algorithm using minimum spanning trees, minimum-weight perfect matching, Eulerian circuits, and metric shortcutting; benchmarked solution quality against exact mixed-integer optimisation solutions.

Manufacturing Systems Capacity and Resource-Allocation Analysis
Python | SQL | Tableau

- Developed quantitative what-if models to evaluate production outcomes under alternative assumptions for efficiency, machine capacity, operating days, production volume, and resource availability.
- Modelled required versus available production minutes to identify capacity gaps and quantify operational implications under alternative efficiency scenarios.
- Applied scenario and resource-allocation analysis to support production-planning decisions and evaluate operational trade-offs.

Data-Driven Manufacturing Decision Support
Python | SQL | scikit-learn | XGBoost | Tableau

- Developed machine-transition and downtime analyses to evaluate production configurations, recurring bottlenecks, production loss, and operational cost impacts.
- Built machine-learning workflows using Random Forest and XGBoost to estimate Standard Minute Value (SMV) for production planning and manufacturing decision support.

Professional Experience

Business Intelligence Analyst June 2024–Present
INDESORE Group *Dhaka, Bangladesh*

- Developed productivity analytics supporting an increase in operator efficiency from **73.9% to 80.8%**; the analysis identified an estimated incremental revenue opportunity of approximately **$1.16 million**.
- Develop quantitative decision-support analytics for production capacity, machine utilisation, efficiency, downtime, Standard Minute Value, and resource allocation in manufacturing operations.
- Build scenario-analysis and performance-monitoring frameworks to identify production gaps, operational constraints, capacity limitations, and resource-allocation opportunities.

Conference and Presentations

[1] **Md Arif Hossain**, Chandrani Nag, "Rubik's Cube – An Application of Group Theory," *International Conference on Science, Technology, Engineering, Mathematics, and Education*, 2023. Presented as a Contributory Talk. [Presentation Certificate]

[2] **Md Arif Hossain**, Chandrani Nag, "The Alternative Form of the Order of an Element in a Group Under Addition Modulo," *International Conference on Science, Technology, Engineering, Mathematics, and Education*, 2023. Presented as a Contributory Talk. [Presentation Certificate]

[3] Sadia Ahmed, **Md Arif Hossain**, Md Anisujjaman, Mohammed Minhaj Uddin, Md. Shafiul Alam Forhad, "Fine-Grained Classification of Depression, Loneliness, and Fear in Bangla Social Media Text," *29th International Conference on Computer and Information Technology (ICCIT 2026).* (Submitted)

Technical Skills

Mathematical Modelling & Optimisation: Gurobi, PuLP, SciPy/HiGHS, NetworkX; mixed-integer programming, combinatorial optimisation, network optimisation, facility location, routing, mathematical programming
Scientific Computing & Programming: Python, MATLAB, R, C++, Fortran, SQL; NumPy, SciPy, pandas
Machine Learning & Data-Driven Modelling: scikit-learn, TensorFlow/Keras, PyTorch, XGBoost; predictive modelling, classification, model development and evaluation
Analytics & Visualisation: Tableau, Power BI, DAX, Excel
Development & Research Tools: Git, GitHub, Jupyter, Google Colab, LaTeX, Streamlit, Flask

Leadership and Additional Achievements

Instructor & Head, Cubing and Puzzle Wing February 2023–March 2024
SUST Science Arena *Shahjalal University of Science and Technology*

- Led the Cubing and Puzzle Wing and organized workshops and competitions for school and university students.
- Mentored students in cubing, puzzles, and logical problem-solving.

Additional Achievements

- **Champion**, Mechnovation 2022 – National Mechanical Festival, MEAS, SUST *October 2022*
- **Top 20 in Bangladesh**, Knights Cubing Open 2019, World Cube Association *September 2019*
- **Champion**, Mechnovation 2019 – National Mechanical Festival, MEAS, SUST *September 2019*

English Proficiency

IELTS Academic: Overall Band 7.0 (Listening 7.5, Reading 7.5, Writing 6.5, Speaking 6.5).